



Name _____ HR _____

8

8th Math Unit 8 - Exponents and Scientific Notation STAR Cards!

QUESTION	ANSWER
8a)  What does 5^4 mean?	8a) $5 \times 5 \times 5 \times 5$ <i>Multiply the base (5) by itself the number of times the exponent (4) tells you</i>
8b) Evaluate:  1) $5^0 =$ 2) $9^0 =$ 3) $4,567^0 =$ 4) What does any number to the zero power equal?	8b) 1) 1 2) 1 3) 1 4) 1

QUESTION	ANSWER																																										
8c)☆☆☆ 1) Fill in the table: <table><tr><td>5⁴</td><td>5 x 5 x 5 x 5 =</td><td>625</td></tr><tr><td>5³</td><td></td><td></td></tr><tr><td>5²</td><td></td><td></td></tr><tr><td>5¹</td><td></td><td></td></tr><tr><td>5⁰</td><td></td><td></td></tr><tr><td>5⁻¹</td><td></td><td></td></tr><tr><td>5⁻²</td><td></td><td></td></tr></table> 2) As the exponents go down by one, what do you notice?	5 ⁴	5 x 5 x 5 x 5 =	625	5 ³			5 ²			5 ¹			5 ⁰			5 ⁻¹			5 ⁻²			8c) As exponents go down by one... <table><tr><td>5⁴</td><td>5 x 5 x 5 x 5 =</td><td>625</td></tr><tr><td>5³</td><td>5 x 5 x 5 =</td><td>125</td></tr><tr><td>5²</td><td>5 x 5 =</td><td>25</td></tr><tr><td>5¹</td><td>5 =</td><td>5</td></tr><tr><td>5⁰</td><td>1 =</td><td>1</td></tr><tr><td>5⁻¹</td><td>$\frac{1}{5} =$</td><td>$\frac{1}{5}$</td></tr><tr><td>5⁻²</td><td>$\frac{1}{5 \times 5} =$</td><td>$\frac{1}{25}$</td></tr></table> ...the answer is divided by the base	5 ⁴	5 x 5 x 5 x 5 =	625	5 ³	5 x 5 x 5 =	125	5 ²	5 x 5 =	25	5 ¹	5 =	5	5 ⁰	1 =	1	5 ⁻¹	$\frac{1}{5} =$	$\frac{1}{5}$	5 ⁻²	$\frac{1}{5 \times 5} =$	$\frac{1}{25}$
5 ⁴	5 x 5 x 5 x 5 =	625																																									
5 ³																																											
5 ²																																											
5 ¹																																											
5 ⁰																																											
5 ⁻¹																																											
5 ⁻²																																											
5 ⁴	5 x 5 x 5 x 5 =	625																																									
5 ³	5 x 5 x 5 =	125																																									
5 ²	5 x 5 =	25																																									
5 ¹	5 =	5																																									
5 ⁰	1 =	1																																									
5 ⁻¹	$\frac{1}{5} =$	$\frac{1}{5}$																																									
5 ⁻²	$\frac{1}{5 \times 5} =$	$\frac{1}{25}$																																									
8d)☆☆☆ Fill in the table below with equivalent values: <table><tr><td>4⁻³</td><td>=</td><td>?</td></tr><tr><td>9⁻⁵</td><td>=</td><td>?</td></tr><tr><td>?</td><td>=</td><td>$\frac{1}{7^4}$</td></tr><tr><td>?</td><td>=</td><td>$\frac{1}{2^8}$</td></tr></table>	4 ⁻³	=	?	9 ⁻⁵	=	?	?	=	$\frac{1}{7^4}$	?	=	$\frac{1}{2^8}$	8d) <table><tr><td>4⁻³</td><td>=</td><td>$\frac{1}{4^3}$</td></tr><tr><td>9⁻⁵</td><td>=</td><td>$\frac{1}{9^5}$</td></tr><tr><td>7⁻⁴</td><td>=</td><td>$\frac{1}{7^4}$</td></tr><tr><td>2⁻⁸</td><td>=</td><td>$\frac{1}{2^8}$</td></tr></table>	4 ⁻³	=	$\frac{1}{4^3}$	9 ⁻⁵	=	$\frac{1}{9^5}$	7 ⁻⁴	=	$\frac{1}{7^4}$	2 ⁻⁸	=	$\frac{1}{2^8}$																		
4 ⁻³	=	?																																									
9 ⁻⁵	=	?																																									
?	=	$\frac{1}{7^4}$																																									
?	=	$\frac{1}{2^8}$																																									
4 ⁻³	=	$\frac{1}{4^3}$																																									
9 ⁻⁵	=	$\frac{1}{9^5}$																																									
7 ⁻⁴	=	$\frac{1}{7^4}$																																									
2 ⁻⁸	=	$\frac{1}{2^8}$																																									

QUESTION	ANSWER
8e) ☆☆☆ 1. Simplify: $10^3 \cdot 10^4$ 2. Finish this sentence: <i>"When multiplying numbers with the same base you..."</i>	8e) 1. $10^3 \cdot 10^4 = 10^7$ 2. When multiplying numbers with the same base you ADD the exponents
8f) ☆☆☆ 1. Simplify: $\frac{10^5}{10^2}$ 2. Finish this sentence: <i>"When dividing numbers with the same base you..."</i>	8f) 1. $\frac{10^5}{10^2} = 10^3$ 2. When dividing numbers with the same base you SUBTRACT the exponents
8g) ☆☆☆ 1. Simplify: $(5^3)^2$ 2. Finish this sentence: <i>"When raising an exponent to an exponent you..."</i>	8g) 1. When raising an exponent to an exponent you MULTIPLY the exponents

QUESTION	ANSWER
8h)  The number below is written in scientific notation. 9.23×10^6 Explain how you would write this number in standard form.	8h) ● Because the exponent of 10 is 6, move the decimal 6 places to the right ● Put a 0 for each empty place value ● The number would be: $9,230,000$
8i)  Write the number below in Scientific Notation. 0.0078	8i) 7.8×10^{-3}